

Workstation Agent & Printer Architecture

This document explains how receipt printing actually gets from the server to a physical printer in this system, how printers are shared across workstations and businesses, and how to set up and troubleshoot it as an admin. It reflects the architecture as of MBM-283 (multi-business-per-agent) and the zombie-printer cleanup that followed it.

1. Core concepts

Term	Meaning
Server	The central app (this Next.js app + database). Runs on one machine, reachable by every workstation on the network.
Business	A tenant in the multi-business app (e.g. "Happy Eater", "Jiffy Lube"). Each has its own printers, users, POS.
Workstation	A physical PC on the local network — a till, a kitchen printer station, an office desktop.
Workstation Agent (<code>WorkstationAgents</code> table)	A <i>pairing</i> between one workstation and one business on one server. Not "one row per PC" — see §3.
Local Agent (<code>r710-agent.exe</code>)	The small background process that runs on a workstation, holds one persistent connection per pairing, and relays print jobs + scale readings.
Network Printer (<code>NetworkPrinters</code> table)	One printer record. Can be DIRECT (server talks to it over IP), AGENT (relayed through a workstation agent), or registered for QZ Tray (a separate, browser-driven path — see §2c).

1a. The three printing paths

This app can print three genuinely different ways. They are not interchangeable and a printer only ever belongs to one of them:

1. **DIRECT mode** — the server itself opens a socket to the printer's IP and prints over the network (ESC/POS raw TCP, typically port 9100). No agent involved. Business-agnostic: a DIRECT printer has no `workstationAgentId` and is not scoped to any business.
2. **AGENT mode** — the server sends the print job over the persistent agent socket to a specific paired workstation, which prints to a Windows-installed printer using the local print spooler. This is what the rest of this document is mostly about. Business-scoped: every AGENT printer belongs to exactly one `WorkstationAgents` pairing, which belongs to exactly one business.
3. **QZ Tray** — a completely separate, browser-driven path. The *browser*, not the server, talks to QZ Tray running on whatever machine the browser tab is open on, which then prints locally. No agent, no server-side routing, no business scoping — it is purely "whatever this browser last saved as its QZ printer." It exists mainly as a zero-agent-setup fallback for a machine with nothing else configured.

Two printers can share the exact same physical hardware and printer name (e.g. "EPSON TM-T") while being registered under two different modes at once — that is not a bug, just two independent ways of reaching the

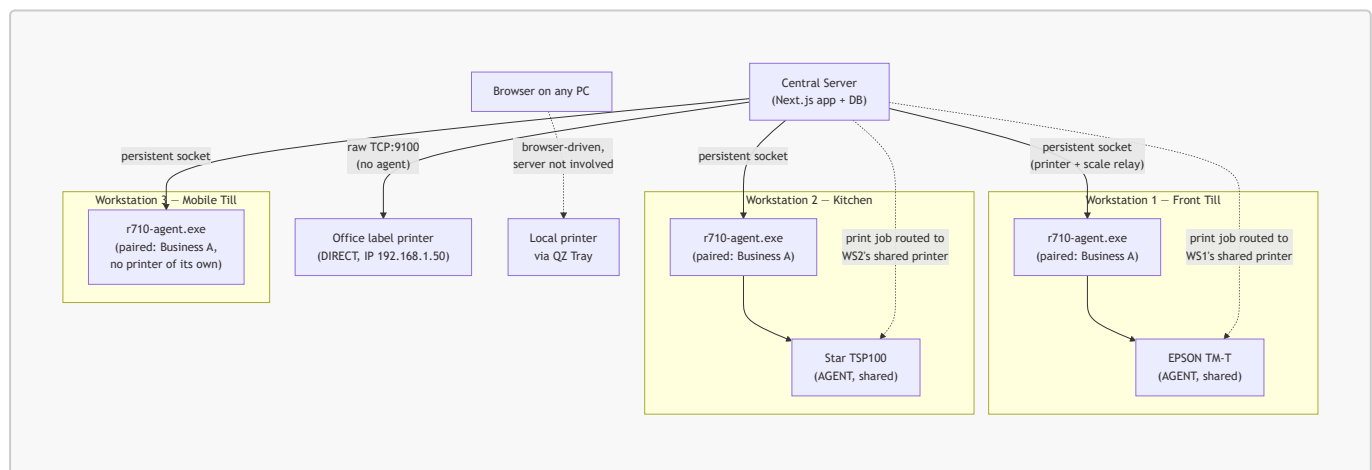
same box.

1b. Key `NetworkPrinters` fields

Field	Meaning
<code>connectionMode</code>	'AGENT' or 'DIRECT'.
<code>workstationAgentId</code>	Which <code>WorkstationAgents</code> pairing this printer is physically attached to. Only set for AGENT mode.
<code>remotePrintingEnabled</code>	"Is this printer reachable from the server at all right now?" Off = paused, not deleted — the printer definition is kept, but no job will ever be routed to it.
<code>remoteEnabled</code>	"Share this printer" — can devices <i>other than</i> the workstation it's attached to route to it? Only meaningful (and only ever true) while <code>remotePrintingEnabled</code> is also true.
<code>isShareable</code>	An older, unrelated flag from before AGENT mode existed. Don't confuse it with <code>remoteEnabled</code> — it's a different field with a different history and doesn't gate AGENT-mode routing.

A workstation can **always** print to its own declared AGENT printer even when `remoteEnabled` is off — "share this printer" only controls whether *other* workstations in the business can also route to it.

2. Architecture diagram



Every arrow from the Server into an agent is a **persistent, always-on socket connection** — not a one-off request. A print job is one self-contained round trip over whichever socket is relevant; there is no "active workstation" exclusivity for printing (unlike the physical scale, which genuinely can only be owned by one business at a time — see §3b).

3. Connection lifecycle

3a. Pairing

Pairing happens once per (workstation, business) combination, from **Admin** → **Workstation Agents**. It mints a token, stores a new `WorkstationAgents` row (`businessId`, `label`, `hostname`, a hash of the token), and the

local agent on that machine saves the credential and connects. From then on the agent reconnects automatically on every restart/reboot — no re-pairing needed unless it's explicitly revoked.

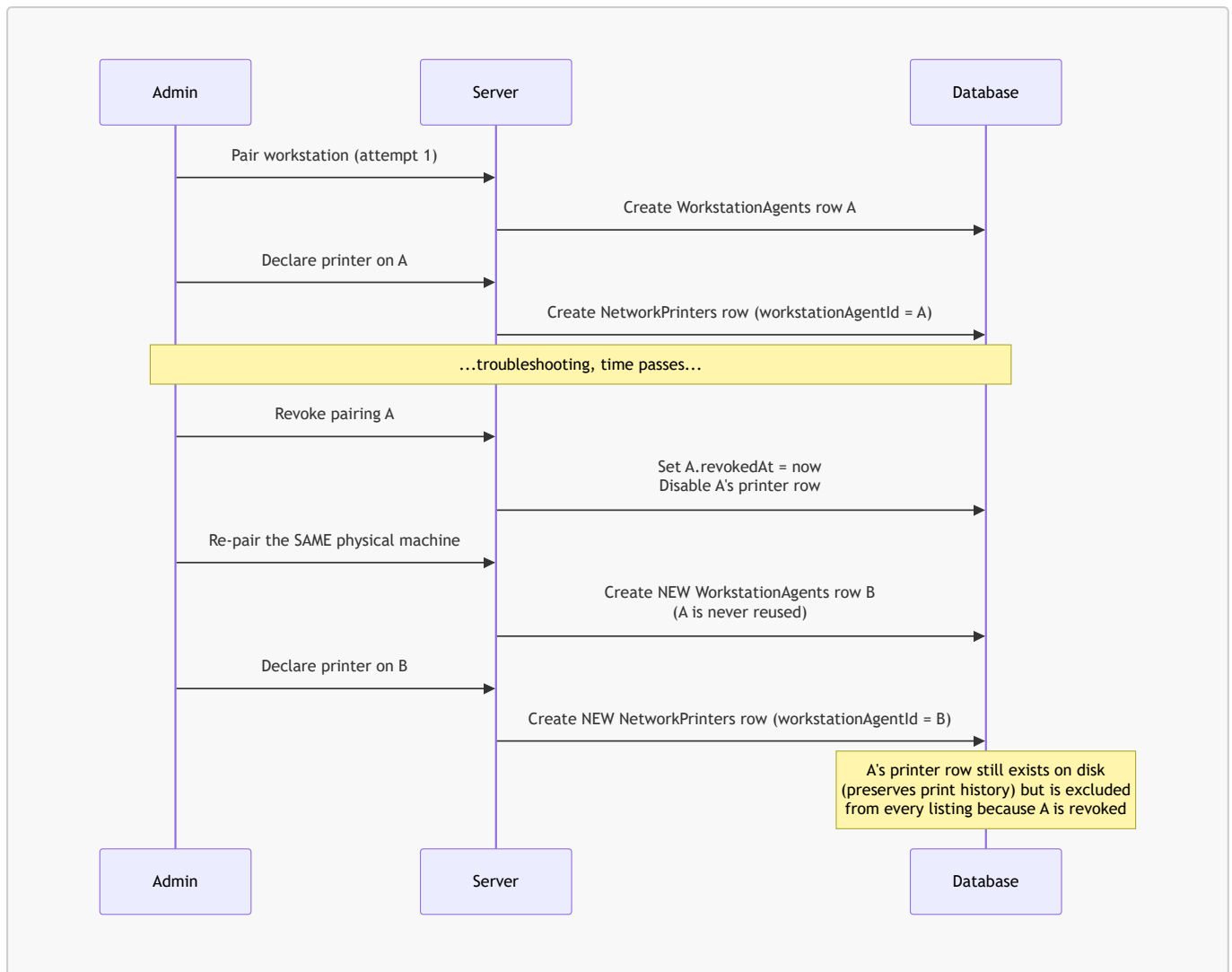
3b. One physical machine, several businesses

The **same physical PC** can be paired to **more than one business** on the same server — e.g. a shared front desk that rings up both "Happy Eater" and "Jiffy Lube." Each pairing is its own independent **WorkstationAgents** row, its own token, and — critically — its own **persistent socket connection**, all running concurrently in the one **r710-agent.exe** process on that machine.

This matters because of a bug this architecture specifically fixes: an earlier design kept only one such socket "active" per physical machine at a time, and switching which business's browser tab had focus would silently disconnect the other business's printer relay. Today, **every paired business's connection is independent and stays up regardless of which business currently has UI focus** — printing has no exclusivity concern at all, since a print job is a single self-contained round trip and the Windows spooler already queues concurrent jobs safely.

The **only** thing that is genuinely exclusive per physical machine is the attached **scale** (one serial port = one hardware resource). Ownership of the scale is arbitrated separately (whichever business currently has browser/tray focus on that machine), and is unrelated to whether either business's printer relay is connected — both stay connected regardless of who currently "owns" the scale.

3c. Revoking a pairing



Revoking (Admin → Workstation Agents → Revoke) sets `revokedAt` on the `WorkstationAgents` row and disconnects its socket. It is a soft, permanent end to that specific pairing — re-pairing the same physical machine for the same business afterward creates a **brand-new** `WorkstationAgents` row, never reuses the old one.

Revoking also:

- Deletes any `ScaleDeviceConfigs` row that referenced the pairing (a stale one otherwise makes the app keep retrying a dead scale connection forever).
- Disables (`remotePrintingEnabled = false`, `remoteEnabled = false`) any `NetworkPrinters` row declared on that pairing — **disabled, not deleted**, because deleting would cascade-delete real print job history (`print_jobs`, `default_receipt_printer_configs` both reference `NetworkPrinters` with `onDelete: Cascade`).

On top of that, every printer *listing* the app does — the print-time picker, the admin printer list — independently excludes an AGENT printer whose owning pairing is revoked, regardless of those flags. This is defense in depth: a pairing revoked before this cleanup existed still self-heals the moment the code runs, with no manual database work needed.

4. Business scoping rules

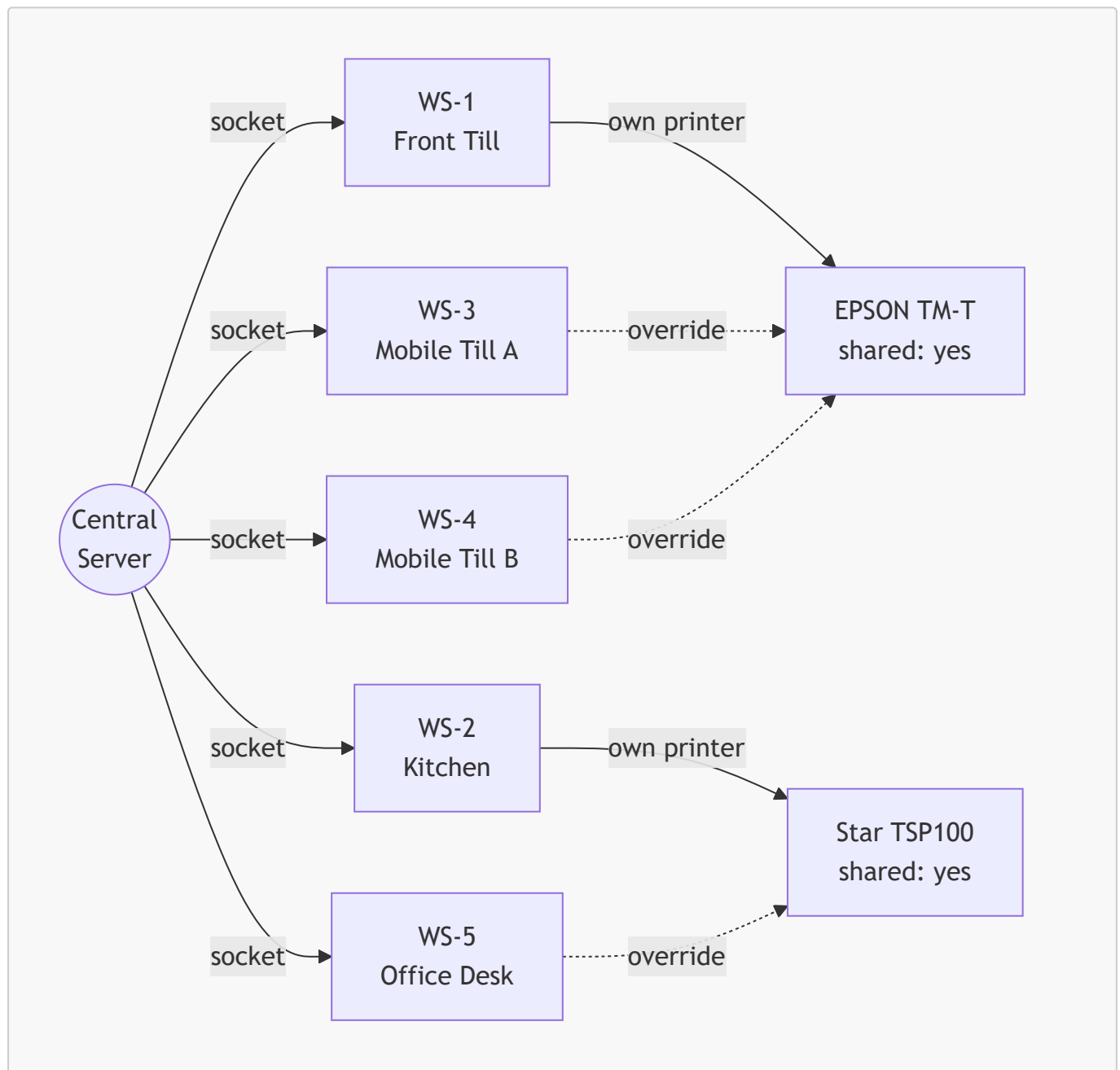
- An **AGENT-mode** printer belongs to exactly one business (via its `WorkstationAgents.businessId`). Only that business's users can print to it, and it only ever appears in that business's printer picker.

- A workstation can **always** discover and print to its own declared AGENT printer, even while `remoteEnabled` (shared) is off.
- A **DIRECT-mode** printer has no business association at all — it's a raw IP the server can reach, available to whichever business's flow happens to be configured to use it. This is true even when the physical machine at that IP is *also* running the server itself — running both roles on one box doesn't change what "DIRECT" means.
- **QZ Tray** printers have no business association either — they're a per-browser, per-machine setting, unrelated to server-side routing entirely.

In short: **business-awareness only exists on the AGENT path.** DIRECT and QZ Tray are both business-agnostic by design, not by omission.

5. Worked scenario — one server, five workstations

Setup: one server, five workstations all paired to the **same business**. Two of the workstations each have their own physical printer, set up for remote printing and shared. The other three have no printer of their own — two of them print through one of the shared printers, the remaining one prints through the other.



Solid arrows are a workstation's own declared printer; dashed arrows are a per-workstation default override routing a printer-less workstation to someone else's shared one.

Workstation	Own printer?	<code>remotePrintingEnabled</code>	<code>remoteEnabled</code> (shared)	Who actually prints here
WS-1 (Front Till)	EPSON TM-T	On	On	WS-1 itself, and WS-3 + WS-4 (see below)
WS-2 (Kitchen)	Star TSP100	On	On	WS-2 itself, and WS-5
WS-3 (Mobile Till A)	none	—	—	Routes to WS-1's EPSON TM-T
WS-4 (Mobile Till B)	none	—	—	Routes to WS-1's EPSON TM-T
WS-5 (Office Desk)	none	—	—	Routes to WS-2's Star TSP100

What's actually configured in the database for this:

- **Two `WorkstationAgents` rows with printers** — WS-1 and WS-2, each with one `NetworkPrinters` row (`connectionMode`: 'AGENT', `workstationAgentId` pointing at that row, `remotePrintingEnabled`: true, `remoteEnabled`: true).
- **Three more `WorkstationAgents` rows with no printer of their own** — WS-3, WS-4, WS-5. They're still paired (so their own socket is live, and they can still receive scale-relay jobs, run local diagnostics, etc.) — they just never declared a physical printer.
- Each of WS-3 and WS-4 has a **per-workstation default printer override** (`DefaultReceiptPrinterConfigs`, keyed by `workstationAgentId`) set to WS-1's printer. WS-5's override is set to WS-2's printer.

What happens at print time on each workstation

The print picker (`UnifiedReceiptPreviewModal`) resolves a default printer in this order (see §6 for the full rule and why):

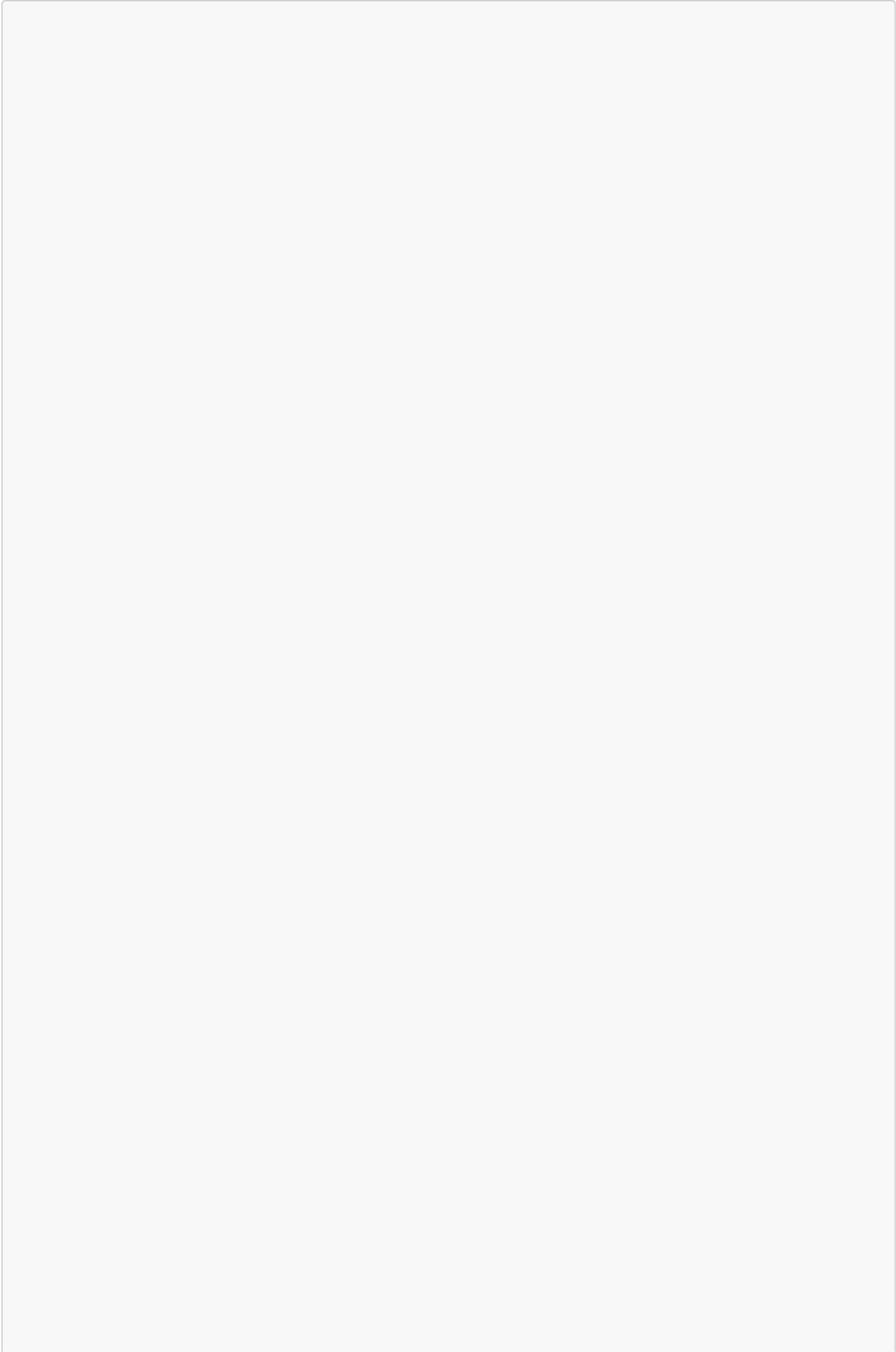
1. This browser's own last-selected printer for *this business* (if still verifiably online).
 2. This workstation's admin-set override, if any.
 3. This workstation's *own* declared AGENT printer, if any.
 4. This business's server-wide default.
 5. QZ Tray's saved printer (last resort, unverified).
- **WS-1 and WS-2**: step 3 resolves immediately — each prints to its own attached printer without any override needed.
 - **WS-3 and WS-4**: no printer of their own, so step 3 finds nothing; step 2's override (set to WS-1's printer) resolves it.
 - **WS-5**: same shape — step 2's override (set to WS-2's printer) resolves it.

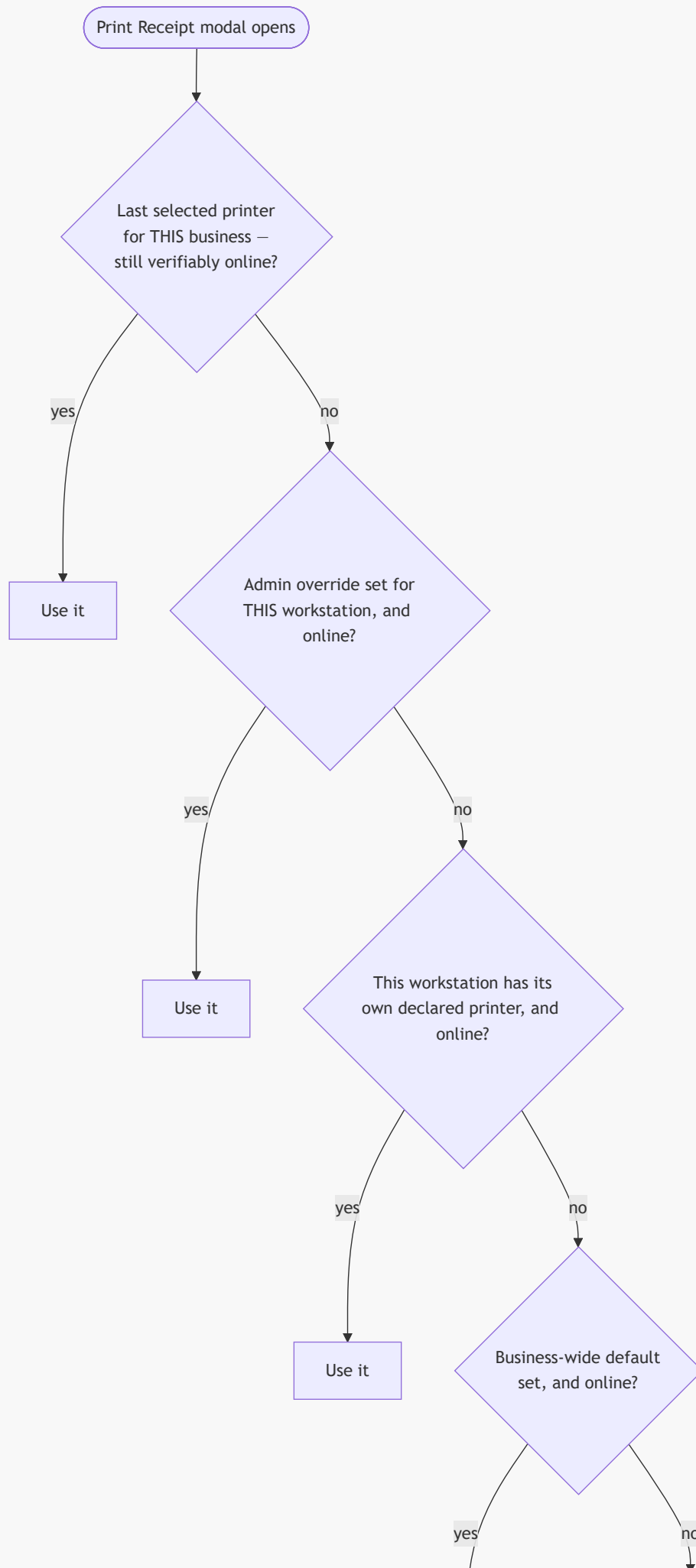
If WS-1's printer goes offline, WS-3/WS-4's override still *points* at it, but the picker's `.isOnline` check skips a dead pick and falls through — worst case, down to the business-wide default (step 4), which an admin should point at whichever shared printer is more reliably up.

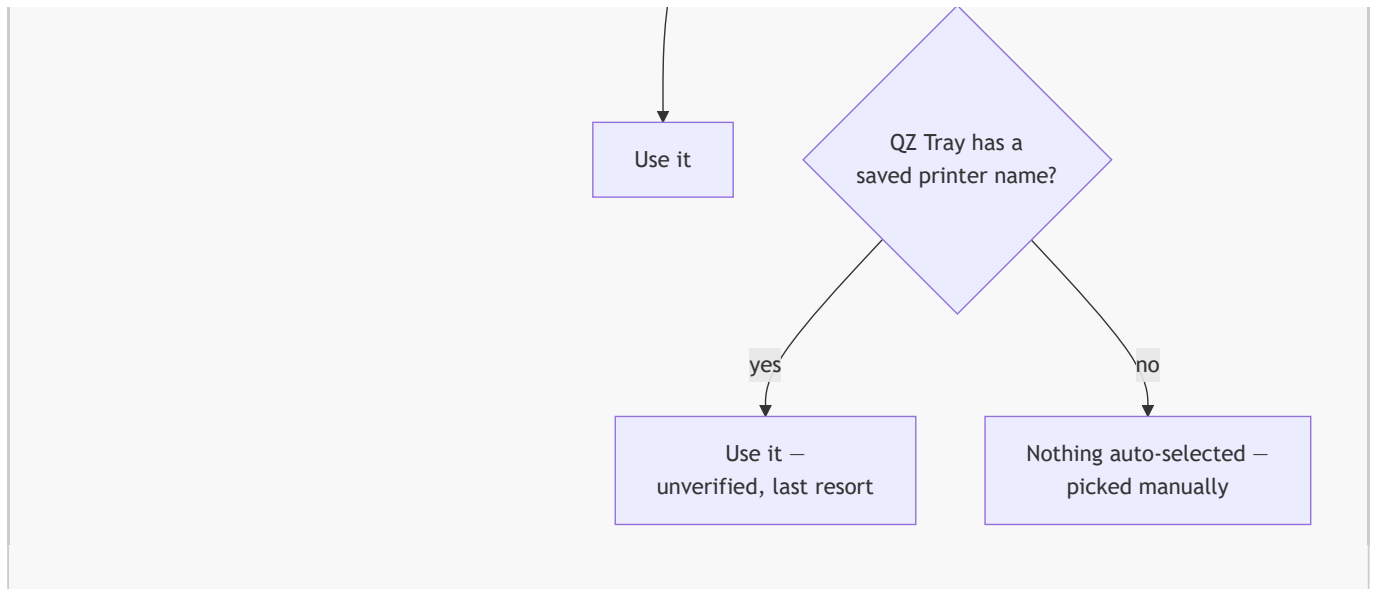
If this were instead spread across two businesses

Nothing above changes conceptually if, say, WS-1/WS-3/WS-4 belong to Business A and WS-2/WS-5 belong to Business B, sharing the same five physical machines and the same server. Each workstation still gets its own independent `WorkstationAgents` row **per business it's paired to** (a machine paired to two businesses has two rows, two tokens, two live sockets — see §3b), and AGENT-mode business-scoping (§4) means Business A's users never see or can route to Business B's printer, even though both printers are physically reachable from the exact same server.

6. Default printer resolution, in detail







Why this order, specifically:

- **Last-selected** wins first because it's the most specific, most recent signal of what a particular person actually wants on a particular machine — but only when it can be *verified* still online. It is scoped per user **and per business** (not globally per user) — a QZ Tray choice made once for one business, or during testing, must never leak into a different business's default.
- **Workstation override** and **own declared printer** both rank above the business-wide default because they're more specific to *this exact machine* — a business default might belong to a completely different workstation.
- **QZ Tray is always last resort**, even when it was the last thing selected, because its "saved printer" is just a remembered *name* — there's no way to verify QZ Tray is actually running without connecting to it (deliberately not done on load, to avoid a permission prompt every time the modal opens). An unverified guess should never outrank something the app can actually confirm is online.

7. Admin setup guide

7a. Pair a new workstation

1. On the target workstation, install and run **r710-agent.exe** (see the download link on **Admin → Workstation Agents**).
2. On that same machine's browser, open **Admin → Workstation Agents** for the business this workstation should serve, and click **Pair**. The local agent's pairing endpoint (<http://127.0.0.1:47710>) is probed automatically — pairing completes with no manual token entry.
3. Repeat step 2, on the same machine, for a **second** business if this workstation should also serve one — this creates a second, independent pairing, not a replacement of the first.

7b. Declare a printer on a workstation

1. On **Admin → Workstation Agents**, find the paired row for the business + workstation in question, click **Set up printer** (or **Edit** if one already exists).
2. Click **List Printers** to pull every printer Windows currently sees on that workstation (the agent must be online), or type the exact name from Windows' own Printers & Scanners settings.
3. Toggle **Enable remote printing** on — this is what makes the printer reachable from the server at all.

4. Toggle **Share this printer** on if other workstations in this business should also be able to route to it. Leave it off if it should only ever be used by this exact workstation.
5. Click **Save**, then click  **Test Print** right there — no need to navigate to a different admin page to confirm it actually works.

7c. Set a business-wide or per-workstation default

- **Business-wide default:** on the same page, use the "Default printer for this business" picker. This is the fallback used by any device with nothing more specific configured.
- **Per-workstation override:** use "Default printer for this workstation" on that workstation's own row. Use this when a workstation with no printer of its own should route somewhere other than the business-wide default (e.g. WS-3/WS-4 in §5's scenario, routed specifically to WS-1's printer rather than whatever the business default happens to be).

8. Troubleshooting playbook

Symptom	Likely cause	Fix
A printer shows "Offline" on a remote workstation but prints fine locally	The printer's owning agent isn't actually connected right now (agent process not running, machine off, network down) — this is a real, live status, not a caching bug.	Check the workstation's tray icon / Admin → Workstation Agents connection status for that pairing. If it says Connected but the printer still shows offline, use Check Status / Bring Online to force a fresh check.
Same printer name appears twice in the picker or admin list, one always offline	A stale "zombie" NetworkPrinters row left over from a revoked pairing that was later re-paired (re-pairing always creates a new WorkstationAgents row, never reuses the old one).	Should self-heal automatically — every listing excludes a printer whose owning agent is revoked. If you still see it, confirm you're on a build that includes this fix; it will not reappear after a fresh page load.
Setting up Business B's printer on a shared workstation made Business A's printer disappear	This was a real historical bug (pre-MBM-283): only one business's connection could be "active" per physical machine at a time.	Fixed — every paired business's connection now stays up simultaneously, regardless of which business currently has UI focus. If this recurs, it's a regression, not expected behavior — check the agent's own version against the "Agent update required" banner.
The wrong printer gets auto-selected at print time (e.g. an unreachable QZ Tray printer instead of a known-online one)	Either a stale per-workstation override pointing at a dead/disabled printer, or a "last selected" browser preference from a different business/earlier testing leaking in.	Confirm the per-workstation and business-wide defaults (§7c) point at a real, online printer. If the picker still won't pick the online one, it's most likely the browser's own remembered choice for <i>this specific business</i> — reselect the correct printer once from the dropdown to overwrite it.

Symptom	Likely cause	Fix
Printer shows "Shared" but a remote workstation still can't reach it	<code>isShareable</code> and <code>remoteEnabled</code> are two different, unrelated flags — the "Shared" badge on the general admin printer list reflects the older <code>isShareable</code> , not whether AGENT-mode remote routing is actually on.	Check Enable remote printing and Share this printer specifically on Admin → Workstation Agents for that printer, not the <code>isShareable</code> badge elsewhere.
Scale readings jump to the wrong business, or a business can't connect its scale	Scale ownership is exclusive per physical machine (one serial port), unlike printers. Whichever business most recently had browser/tray focus on that machine owns it.	Use the tray's Release action (or the equivalent admin control) to hand the scale back explicitly, then refocus the correct business's tab.
"List Printers" on the setup page returns nothing	The agent must be online for this to work — it runs <code>Get-Printer</code> locally on that exact workstation.	Confirm the agent's connection status is green first. If it's online and still returns nothing, type the printer name manually from Windows' own Printers & Scanners settings instead — this always works regardless of whether listing does.

9. Appendix — relevant code

Concern	File
Printer listing / business-scoping / revoked-agent exclusion	<code>src/lib/printing/printer-service.ts</code>
Print-time authorization (is this printer usable by this business right now)	<code>src/lib/printing/print-dispatch.ts</code>
Default-printer resolution priority	<code>src/components/receipts/unified-receipt-preview-modal.tsx</code>
Business-wide / per-workstation default storage	<code>src/app/api/printing/default-printer/route.ts</code>
Pairing, printer declaration, revoke (with cleanup)	<code>src/app/api/admin/workstation-agents/**</code>
Local agent — multi-business persistent connections	<code>agent/r710-local-agent/src/index.ts</code>
Local agent — scale ownership arbitration	<code>agent/r710-local-agent/src/scale-owner.ts</code>